CHAPTER 17

ROUTINE INTERPLANETARY VIOLENCE

It is a law of nature that Earth and all other bodies should remain in their proper places and be moved from them only by violence.

—ARISTOTLE (384–322 B.C.), *PHYSICS*

There was something funny about Saturn. When, in 1610, Galileo used the world's first astronomical telescope to view the planet—then the most distant world known—he found two appendages, one on either side. He likened them to "handles." Other astronomers called them "ears." The Cosmos holds many wonders, but a planet with jug ears is dismaying. Galileo went to his grave with this bizarre matter unresolved.

As the years passed, observers found the ears … well, waxing and waning. Eventually, it became clear that what Galileo had discovered was an extremely thin ring that surrounds Saturn at its equator but touches it nowhere. In some years, because of the changing orbital positions of Earth and Saturn, the ring had been seen edge-on and, because of its thinness, it seemed to disappear. In other years, it had been viewed more face-on, and the "ears" grew bigger. But what does it mean that there's a ring around Saturn? A thin, flat, solid plate with a hole cut out for the planet to fit into? Where does *that* come from?